Technical Research Documentation for the Modular Flexible Bio-Panel Integrated Digester (MFBP-ID) System

This document provides a detailed technical description of the Modular Flexible Bio-Panel Integrated Digester (MFBP-ID) system. The system is designed as a household-scale, closed-loop solution for managing organic kitchen waste and greywater, converting them into useful resources: biogas, clean water, and bio-fertilizer.

1. System Overview

The MFBP-ID system is an integrated, multi-stage process that leverages biological and photobiological principles. It consists of an anaerobic digestion unit for biogas production, a hydrogen sulfide (H2S) scrubber for gas purification, and a series of flexible bio-panels (photobioreactors) for greywater treatment and CO2 fixation. The entire process is designed with a DIY-aesthetic, emphasizing modularity, local materials, and solar-powered components for operational independence.

2. Component-by-Component Analysis

2.1 The Anaerobic Digester (AD) Unit

- **Function:** The AD unit acts as the primary processor for organic household waste. It is a sealed vessel, typically a 200-500 liter polyethylene tank, designed to maintain stable mesophilic conditions.
- **Process:** Organic waste is added through a sealed inlet, where a consortium of anaerobic bacteria ferments the material. This biological process yields two primary products: a gaseous mixture known as biogas and a nutrient-rich liquid digestate.
- **Outputs:** The biogas is collected from a port at the top of the vessel, while the digestate can be periodically drained from a valve at the base.

2.2 The Hydrogen Sulphide (H2S) Scrubber

- **Function:** Positioned downstream of the AD unit, the H2S scrubber is a critical gas purification component. Its purpose is to remove corrosive H2S from the biogas stream.
- **Mechanism:** The scrubber is a small, sealed chamber packed with iron filings or steel wool. The H2 S gas reacts with the iron medium, forming a stable sulfide compound, thus purifying the biogas and preventing corrosion of downstream components and inhibiting algal growth.

2.3 The Flexible Bio-Panels (Photobioreactors)

- **Function:** The bio-panels are the central component for greywater remediation and carbon capture. These are modular photobioreactors (PBRs) designed for high surface area and efficient solar exposure.
- **Structure:** Each panel consists of a simple frame (e.g., bamboo or PVC) supporting a transparent, heavy-gauge plastic sheet. This plastic is sealed to form thin, vertical channels, creating a large, illuminated surface area.
- **Process:** Greywater from the household is introduced at the top of the panels and flows downward through the channels. The greywater carries a culture of microalgae, which, in the presence of sunlight, consume the nutrients (e.g., nitrogen, phosphorus) from the wastewater. This process both purifies the water and generates algal biomass.

2.4 The Integrated Airlift-Style Gassing Troughs

- **Function:** These troughs, located at the base of each bio-panels, are responsible for efficient gas exchange and culture circulation.
- **Mechanism:** Purified biogas from the $H2S$ scrubber is injected into the troughs. The $CO2$ in the biogas provides the necessary carbon source for the microalgae. The injection of gas creates an airlift effect, generating a gentle, upward current that circulates the algal culture throughout the panel, ensuring uniform exposure to sunlight and nutrients.

2.5 Pumping, Piping, and Solar Power

- **Circulatory System:** The system's "circulatory" network comprises low-power, solar-driven pumps and a network of PVC pipes and flexible hoses. A solar-powered pump moves pre-treated greywater from a holding tank to the top of the bio-panels. A second small air pump injects the biogas into the gassing troughs.
- **Piping:** A dedicated network of pipes connects all components, including a separate line for the methane-rich biogas to a cooking stove, and another for the treated water to a reservoir for reuse.
- **Energy Source:** A small solar photovoltaic (PV) system, including a panel and a battery (UPS), provides the necessary power for the system's pumps, ensuring energy independence.

3. Process Flow and Integration

The MFBP-ID system operates as a continuous, integrated cycle:

1. Organic kitchen waste is fed into the AD unit, producing biogas and digestate.
2. Biogas is routed through the $H2 S$ scrubber for purification.

3. The purified biogas is then split: the methane-rich portion is directed for cooking, while the CO_2 is injected into the bio-panel troughs.
4. Greywater is pre-treated and pumped to the top of the bio-panels.
5. The greywater flows down the panels, where algae consume nutrients and the injected CO_2 under solar illumination.
6. The treated water is collected in a final reservoir for reuse, and the nutrient-rich algal biomass is periodically harvested as a potent bio-fertilizer.

4. Alternative Simplified Design: Atmospheric Carbon Capture

An alternative, simpler design is proposed to provide a more streamlined and lower-cost greywater treatment solution. This design removes the complexity of the anaerobic digestion process while maintaining the core principles of algal greywater remediation.

- **Key Difference:** The primary distinction of this design is the removal of the anaerobic digester and the H_2S scrubber. Instead of utilizing biogas as a source of CO_2, this system relies on a process of atmospheric carbon capture.
- **Process:** An air pump, powered by the same solar PV system, draws ambient air and injects it directly into the airlift-style gassing troughs at the base of the bio-panels.
- **Mechanism:** The microalgae in the greywater culture absorb the atmospheric CO_2 that is dissolved in the water from the injected air bubbles. While the concentration of CO_2 is lower than in biogas, the continuous supply from the air pump is sufficient to support algal growth and the associated greywater treatment.
- **Benefits:** This simplified model reduces the number of components, the overall footprint, and the initial capital investment. It is ideal for applications where biogas for cooking is not a primary objective, and the focus is solely on sustainable greywater treatment and bio-fertilizer production

Modular Greywater Biogas and Algae Treatment System Report

1: Introduction

This report details the technical design, specifications, calculations, and key assumptions for a Modular Greywater Biogas and Algae Treatment System designed for a 4-person household in Nasik, Maharashtra, India.

The system integrates anaerobic digestion of kitchen waste producing biogas, H_2S scrubbing using iron oxide media, and algal cultivation for greywater treatment and CO_2 capture. The design focuses on sustainability, cost-effectiveness, and practical implementation.

1. Anaerobic Digester Outputs

Initial Input: Total Daily Biogas Volume: 180 Liters/day Assumed Biogas Composition:

- Methane (CH_4): 60% (v/v)
- Carbon Dioxide (CO_2): 35% (v/v)

- Hydrogen Sulphide (H_2S): 0.5% (v/v)

Calculations:

- Daily CO_2 Volume from Biogas: 180 L/day×0.35=63 L/day

- Daily H_2S Volume from Biogas: 180 L/day×0.005=0.9 L/day

- Daily H_2S Mass from Biogas: 0.9 L/day×1.36 g/L=1.224 grams/day

II. H_2S **Scrubber Size**

Key Inputs and Assumptions:

- Daily H_2S Mass from Biogas: 0.001224 kg/day

- Desired Media Lifespan: 182.5 days (6 months)
- Iron Oxide Media Capacity: 0.28 kg H_2S per kg of media (Illustrative assumption)
- Safety Factor: 1.3

Calculations:

- Total H_2S to be Removed over Desired Lifespan: 0.001224 kg/day×182.5 days = 0.22386 kg H_2S
- Theoretical Mass of Iron Oxide Media Required: 0.22386 kg/0.28=0.7995 kg media
- Actual Mass of Iron Oxide Media (with Safety Factor): 0.7995×1.3=1.039 kg media
- Volume of Media Required: 1.039 kg/800 kg/m^3=0.00130 m^3 (1.30 Liters)
- Conceptual Scrubber Dimensions:
  - Diameter: 15 cm (0.15 m)
  - Height: 7.35 cm

4: Dimensions of Panels

III. Dimensions of Panels

- Required Volume per Panel: 936 Liters/3 panels=312 Liters/panel

- Dimensions per Panel:
  - Height (H): 2.0 m
  - Depth (D): 0.15 m
  - Width (W): Volume / (H×D)=0.312 m^3/(2.0 m×0.15 m)=1.04 m

- Summary:
  - 3 Panels each 1.04 m wide x 2.0 m high x 0.15 m deep
  - Total System Volume = 936 Liters

IV. Greywater Treatment Capacity & Algal Biomass Production

Key Assumptions:

- Daily Greywater Flow (Q): 144 L/day (4-person household)
- Influent Phosphorus (P_{in}): 0.330 mg/L (Total P)
- Influent Nitrogen (N_{in}): 18.98 mg/L (Total N)
- Target Removal Efficiency: 95%
- Hydraulic Retention Time (HRT): 6.5 days
- Algae N Content (dry biomass): 8%
- Algae P Content (dry biomass): 1%
- Algae Carbon Content (dry biomass): 50%
- Biogas CO_2 Content: 35%
- CO_2 Density (STP): 1.98 g/L

Calculations:

- Bioreactor Volume: 144 L/day×6.5 days=936 Liters

- Mass of P to Remove Daily: 0.330 mg/L×144 L/day×0.95=45.144 mg P/day

- Mass of N to Remove Daily: 18.98 mg/L×144 L/day×0.95=2594.16 mg N/day

- Daily Dry Biomass for P Removal: 45.144 mg/0.01=4514.4 mg/day

- Daily Dry Biomass for N Removal: 2594.16 mg/0.08=32427 mg/day

- Final Daily Dry Biomass Production: 0.03243 kg/day

- Daily CO_2 Volume from Biogas: 180 L/day×0.35=63 L/day

- Daily CO_2 Mass from Biogas: 63 L/day×1.98 g/L=124.74 g/day (0.12474 kg/day)
- Mass of Carbon in Biomass: 0.03243 kg/day×0.50=0.016215 kg C/day

- Mass of CO_2 Captured: 0.016215×(44/12)=0.059455 kg CO_2 /day

3. Conclusion and Recommendations

The system presents a viable solution for reducing household environmental footprint by producing clean water and renewable energy.

Further analysis needed includes:

- Detailed greywater organic content analysis for accurate biogas yield

- Economic payback and cost-benefit analysis
- Electrical power and solar panel sizing
- Evaluation of CO_2 dilution and H_2S scrubber efficiency under real-world conditions

4. Key Assumptions and Citable Sources

Biogas Production and Composition:

- The assumed daily biogas volume of 180 Liters/day is a conservative estimate based on the performance of similar household-scale anaerobic digestion systems in India. For example, the Appropriate Rural Technology Institute (ARTI) has developed a compact plant for households in Maharashtra that processes 1-2 kg of food waste per day. Furthermore, systems developed by BIOTECH in Kerala, South India, demonstrate a yield of approximately 1 cubic meter (1000 Liters) of biogas from 5 kg of kitchen waste for a family of 3-5 people, providing a contextual benchmark for the calculations.
- The assumed biogas composition of 60% methane (CH_4), 35% carbon dioxide (CO_2), and trace amounts of hydrogen sulphide (H_2S) is supported by academic literature. In the study (Tippayawong & Thanompongchart, 2010), it was found that Typical biogas contains 50–65% methane (CH_4), 30–45% carbon dioxide (CO_2), moisture and traces of hydrogen sulphide (H_2S).

Solar Insolation for Nasik, India:

- India Meteorological Department (IMD) Climate Data, 2023. https://mausam.imd.gov.in/

  Pump Power and System Losses:
- Ugwu, C. U., Aoyagi, H., & Uchiyama, H. (2008). Photobioreactors for mass cultivation of algae. Bioresource Technology, 99(10), 4021-4028. DOI:10.1016/j.biortech.2007.01.046

Assumptions Summary:

- Daily Greywater Volume: 144 L/day (for a 4-person household).
- Hydraulic Retention Time (HRT): 6.5 days.
- Influent Phosphorus (P_{in}): 0.330 mg/L (Total P).
- Influent Nitrogen (N_{in}): 18.98 mg/L (Total N).
- Ammonium (NH_4^+-N) is a significant fraction of Total Nitrogen.
- Daily Biogas Volume: 180 L/day.
- Biogas Composition: 60% CH_4, 35% CO_2, and 0.5% H_2S.
- Steady-state operation, constant greywater input, and temperature $25-30^\circ C$ (typical Nasik climate).

- 95% nutrient removal efficiency.
- CO_2 dilution to 15% before algal introduction.
- 90% H_2S scrubber efficiency.
- System designed for a typical 4-person household in Nasik, Maharashtra, India.